

Effectiveness of Structured Breathwork Interventions on Heart Rate Variability and Heart Rate Recovery Following Maximal Exertion in Tactical Athletes

MARK E. JONES¹, GENEVIEVE SMITH², ANTHONY ACEVEDO³ and BRIDGET MELTON^{1,4}

¹Department of Health and Human Performance, Concordia University of Chicago, River Forest, IL; ²Defense Health Agency Public Health—Aberdeen, Aberdeen Proving Ground, MD; ³College of Natural Sciences, Grand Canyon University, Phoenix, AZ; and ⁴Department of Health Sciences and Kinesiology, Georgia Southern University, Statesboro, GA

ABSTRACT

JONES M. E., G. SMITH, A. ACEVEDO, and B. MELTON. Effectiveness of Structured Breathwork Interventions on Heart Rate Variability and Heart Rate Recovery Following Maximal Exertion in Tactical Athletes. *Med. Sci. Sports Exerc.*, Vol. 58, No. 7, pp. 1524–1533, 2026. **Purpose:** This study investigated the acute effects of two structured breathwork methods on autonomic recovery following maximal exertion. **Materials and Methods:** A randomized control trial was conducted with 62 cadets (44 males and 18 females) assigned to three distinct groups: box breathing (BB), cyclic sighing (CS), or spontaneous breathing control. Participants performed a graded exercise test to assess maximal oxygen consumption (VO_2). High-frequency heart rate variability (HF-HRV) and heart rate recovery (HRR) were collected to evaluate autonomic recovery and parasympathetic reactivation. HF-HRV was analyzed at three time points: baseline (Pre), 1–3-min postexercise recovery (Post1), and 4–6-min postexercise (Post2). HRR was measured at 1- and 3-min intervals, analyzing reductions from peak heart rate captured at $\text{VO}_{2\text{max}}$. All analyses were evaluated using a 3 (group) $\times$ 3 (time) mixed-design repeated-measures analysis of variance. **Results:** All groups were comparable in age, weight, height, and body mass index (21 ± 3 yr; 77.4 ± 15.4 kg; 173.1 ± 9.0 cm; 25.7 ± 4.0 kg·m⁻², respectively). HF-HRV revealed significant main effects for time ($F_{(2,118)} = 465.01$, $P < 0.001$, $\eta^2=0.89$), group ($F_{(2,59)} = 8.20$, $P < 0.001$, $\eta^2=0.22$), and time $\times$ group interaction ($F_{(4,118)} = 3.54$, $P = 0.009$, $\eta^2=0.21$). Both the CS and BB groups had greater increases in HF-HRV from Post1 to Post2 when compared to spontaneous breathing ($P = 0.001$). HRR improved across all participants from 1- to 3-min postexercise with n group level differences ($F_{(1,59)} = 33.51$, $P < 0.001$, $\eta^2=0.36$). **Conclusions:** Structured breathwork following maximal exertion significantly accelerated parasympathetic reactivation as reflected by HF-HRV. Both BB and CS offer a simple field-ready intervention aimed to enhance operational readiness for tactical populations through enhancing acute physiological recovery. **Key Words:** BOX BREATHING, CYCLIC SIGHING, RESERVE OFFICER TRAINING COMMAND

Operational readiness is the ability to mobilize and meet the demands of assigned missions (1) and maintain multidimensional functionality of physical and psychological fitness when called upon (2). “Tactically effective” is a more nuanced term regarding the ability to win small battles, or avoid defeat through engagements, skirmishes, or strategic maneuvers, and

extends into law enforcement and fire service in similar manners, encompassing mental resilience and physical capabilities (3). Just as military forces prepare for mission execution, both police and fire units must maintain an operationally ready status to respond effectively under pressure and make decisions under stress. Therefore, tactical effectiveness translates to the ability to apply training, master communication, and control volatile incidents while protecting the lives of citizens. Tactical personnel often encounter physically demanding scenarios and tasks requiring a rapid shift in cardiovascular recovery to sustain operational readiness and remain tactically effective.

Following maximal physical exertion, the autonomic nervous system must quickly shift from a sympathetic dominance to parasympathetic reactivation. This transition is vital for improving decision-making, reducing fatigue, and preparing for subsequent high-intensity scenarios. Heart rate variability (HRV) and heart rate recovery (HRR) are well-established markers of autonomic function, with higher HRV and quicker HRR

Address for correspondence: Mark E. Jones, Ph.D., Department of Health and Human Performance, Concordia University of Chicago, River Forest, IL 60305; E-mail: Crf_jonesme@cuchicago.edu; The-startingpoint3@gmail.com

Submitted for publication December 2025

Accepted for publication February 2026

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.acsm-msse.org).

0195-9131/2026/587-1524/1533

MEDICINE & SCIENCE IN SPORTS & EXERCISE®

Copyright © 2026 by the American College of Sports Medicine

DOI: 10.1249/MSS.0000000000003972

indicating physiological recovery (4–6). High-frequency HRV (HF-HRV), the spectral component ranging from 0.15 to 0.40 Hz, reflects vagally mediated parasympathetic activity and is considered a sensitive indicator of neurocardiac recovery, specifically after high stress or physical exertion (5). HRR, known to be a prognostic marker of cardiovascular health, also serves as an acute measure of resilience and parasympathetic reactivation. Faster HRR following maximal exertion demonstrates a superior ability to control the responsiveness of the autonomic nervous system and the efficiency of the cardiovascular system (7,8). Conversely, delayed HRR has been associated with increased fatigue accumulation and autonomic imbalance; ultimately resulting in an increased risk of long-term cardiometabolic dysfunction (7–9). For tactical personnel, sustaining rapid HRR across training cycles signifies improved readiness and recovery capacity (8,10). Therefore, optimizing physiological recovery capability through monitoring HRV and HRR may have a direct positive influence on health and performance outcomes in tactical populations, which are key components of operational success and the preservation of long-term health.

Structured breathwork may help influence autonomic regulation, especially through techniques that enhance vagal tone (11–13). Cyclic sighing (CS), which emphasizes prolonged exhalation, has been shown to reduce physiological arousal, elevate parasympathetic activity, and balance HRV in civilian populations (14). In tactical populations, CS is relevant due to its ability to accelerate autonomic recovery following high-risk scenarios or peak exertion. By promoting vagal reactivation, CS can help tactical populations maintain composure and potentially enhance decision-making capabilities under operational stress, as a byproduct of preventing cognitive fatigue (11,14).

Box breathing (BB), commonly known as tactical breathing, involves an equated inhalation, pause-hold, exhalation, and pause-hold phases. This structured breathwork approach has been shown to be efficacious in stress regulation and maintaining composure within high-pressure special operations populations (11,15,16). Despite emerging support regarding the physiological benefits surrounding breathwork techniques, including autonomic regulation (11,12), neurocognitive stability (16), and metabolic modulation (13,17), applications in tactical populations, specifically following maximal exertion, remain underexplored.

This study evaluates the acute effects of CS and BB on physiological recovery, measured by HF-HRV and HRR, following maximal exertion achieved during a graded exercise test (GXT) in Reserve Officer Training Command (ROTC) cadets. ROTC cadets represent an ideal population for investigating recovery dynamics emerging in the tactical space, as they are currently

undergoing structured military physical training and education, while also still maintaining a civilian status since they are not yet active in the service. This transitional status offers a controlled environment to examine readiness-related adaptations that have previously been studied in civilian populations and translate into tactical contexts. This study addresses a gap in tactical recovery by investigating whether acute structured breathwork interventions can accelerate parasympathetic reactivation compared to spontaneous breathing. It is hypothesized that both CS and BB will significantly enhance recovery and autonomic regulation, as indicated by increasing HF-HRV and a more rapid decline in heart rate, following maximal exertion as compared to spontaneous breathing.

METHODS

Participants

ROTC cadets ($n = 77$, 53 males and 24 females, aged 20.77 ± 2.75 yr) were recruited to participate in a randomized control trial with a mixed repeated-measures design at a midsized southeastern public university. The sex distribution of the present sample is a slightly higher proportion of female participants (29%), whereas female cadets typically represent approximately 20%–26% of commissioned officers in training (18). Of these, 10 participants were excluded due to technical issues with the connectivity of the heart rate straps, resulting in unreadable data during the data collection. An additional five participants were removed during data screening because their HF-HRV values exceeded $15,000 \text{ ms}^2$, representing physiologically implausible outliers exceeding ± 3 standard deviation criterion. Inclusion criteria included active ROTC status, absence of cardiovascular, metabolic, or pulmonary conditions, or a medical clearance if any previous concerns were present with the candidate. Individuals who were pregnant, recently injured, or on a military medical profile restricting physical exertion were excluded. Participation in the study was voluntary and not directly influenced by ROTC leadership. Before beginning the study, cadets were informed that refusal to participate or withdrawal from the study would have no impact on their standing within the ROTC program, and they could discontinue participation at any time. All participants provided informed consent, and the study obtained approval from the appropriate university's Institutional Review Board (H26016) and was further approved by the U.S. Army Cadet Command and the Department of Defense Human Research Protections Office. All research was performed following ethical standards of the Declaration of Helsinki and DoD Instruction 3216.02, "Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research."

Randomization and Blinding

Participants were randomly assigned to one of three intervention groups: BB, CS, or spontaneous breathing (control). Randomization occurred using a computer-generated sequence in R (version 2025.05.1+513). A block design of six was used to ensure balanced group allocation throughout recruitment. Randomization was performed by the principal investigator before data collection and concealed electronically throughout the duration of the study. Due to the nature of the intervention, participants were not blinded throughout the entirety of the investigation. However, all participants were blinded to their interventional groups through the end of the baseline breathing measurement to ensure that baseline HRV metrics were not influenced by breathing treatment training.

Protocol

Upon arrival, participants completed a standard physical activity readiness questionnaire for everyone (PAR-Q+), health history questionnaire, consent, and assent forms for those designated as minors within the ROTC program. PAR-Q+ and health history questionnaire have demonstrated strong test-retest validity and reliability for screenings and risk stratification (19) and have been frequently published by the American College of Sports Medicine in their guidelines for exercise testing and prescription (20). After intake questionnaires, participants were fitted with a Polar H10 chest strap (Polar Electro, Kempele, Finland) to record heart rate and R-R intervals. Krummenacher et al. (21) validated the Polar H10 as a reliable field-based device for HRV, showing strong agreement with electrocardiogram (ECG) recordings. These devices record interbeat intervals at 1000 Hz with minimal artifact when properly positioned. Placement at the lower sternum, maintaining full skin contact and moisture for optimal signal quality, minimizes motion artifacts and ensures stable conductivity, which is vital for HRV measurements in the short-term during both resting and exercising conditions. The Polar H10 has been validated as a reliable device for measurements of R-R intervals and has shown agreement with ECG during both rest and high-intensity training at 99.4%–99.6% (22). Similar approaches with Polar H10 and ECG reported strong agreements in pre- and postexercise conditions, further supporting R-R interval accuracy (23).

Following intake and fitting for VO₂ masks and Polar H10 straps, a 3-min supine HRV measurement was recorded for each participant during spontaneous breathing with zero coaching to establish baseline autonomic conditions. Measurements were taken using Kubios HRV Premium software (Kubios Oy, Kuopio, Finland), which has been validated for short-term autonomic function when paired with Polar H10 devices.

Short-term autonomic assessments demonstrate excellent concurrent validity with ECG-derived R-R intervals, as well as high interrater and intrarater reliability for HRV computation (23,24). Participants then completed a GXT on a Concept2 Model D rower (Concept2 Inc., Morrisville, VT). These ergometers have been validated and deemed reproducible for assessing aerobic fitness in both Olympic style and traditional rowing practices and produced near-perfect correlation with peak VO₂ performance, confirming their reliability during incremental testing (25,26). The rower was calibrated before each test to ensure accuracy with a drag factor set at 135 for female participants and 145 for males. The test began at 100W for males and 60W for females, increasing in 3-min stages by 30W until one of the following conditions was met: volitional exhaustion, defined as the point at which the participant could no longer maintain the work rate specified within the protocol, or self-terminated due to self-reported maximal effort. The rate of perceived exertion at the end of the test was used to confirm exhaustion (rate of perceived exertion = 7.8 ± 1.7) on the BORG CR10 Scale. Additionally, the test was terminated if the participant failed to maintain the 10 o'clock–2 o'clock torso position. These termination criteria followed guidance from established rowing protocols (26,27). Maximal oxygen consumption was obtained through the VO₂ Master Pro (VO₂ Master Health Sensors Inc, Vernon, BC, Canada), version 1.6.1. Montoye et al. (28) reported strong correlations ($r = 0.93$ – 0.97) between ParvoMedics TrueOne 2400 and the VO₂ Master. In their validation study, absolute mean differences were <10% for VO₂ and coefficients ranged from 11.4% to 15.1%, supporting the use of the VO₂ Master Pro as a reliable and valid field-based system. Additionally, Penichet-Tomás et al., (27) validated the VO₂ Master Pro during traditional rowing utilizing the Concept2 Model D rower.

Immediately following test termination, participants engaged in their respective breathing condition for a 3-min cooldown at 50% of their maximal rowing wattage. BB consisted of a 4-s inhalation, a 4-s inspiratory hold, a 4-s exhalation, and a 4-s expiratory hold, performed at approximately 4 breaths per min (11,29). CS involved a single deep inhalation, followed by a second shorter inhalation and a prolonged exhalation, repeated at approximately 5.5 breaths per min (30). Both breathwork protocols are presented in Figure 1. As breathwork techniques were not common practice for many participants, those assigned to the breathwork conditions were advised to gradually progress toward the target cadence. Specifically, participants in the BB group were instructed to work up to the full 4-4-4-4 pattern, as sustaining a 4-s box pattern may initially feel challenging. Nonetheless, all participants were coached in the cool down phase to maintain the respiratory rate of ~4 breaths per min for BB and ~5.5 breaths per min for CS. During this cool-down,

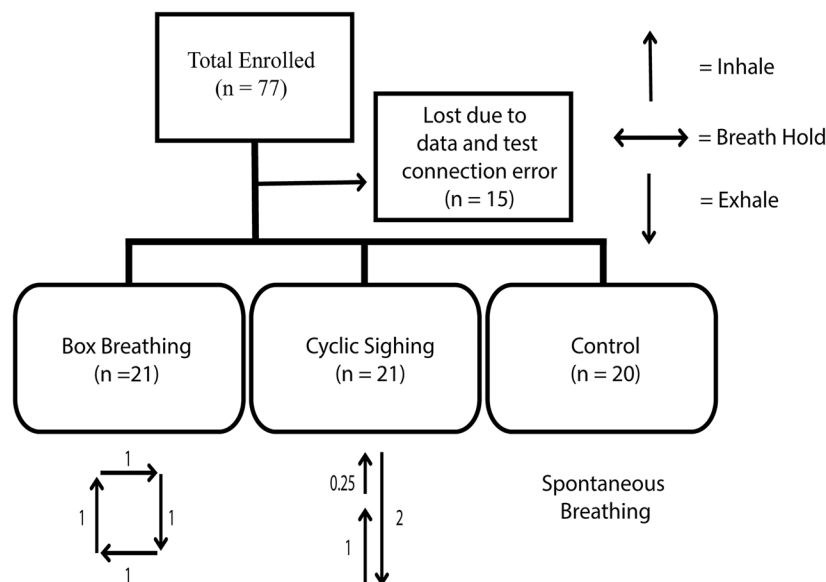


FIGURE 1.—Participant flow and experimental design. Flow diagram depicting recruitment, group allocation, and attrition.

participants in the breathwork conditions received pacing guidance verbally for the first 60 s, during which the principal investigator coached them through approximately 4–5 guided breath cycles to establish rhythm consistency. Following the brief familiarization, participants continued independently while being visually monitored to ensure prescribed pacing adherence. Participants in the control (CTL) group maintained spontaneous breathing during the 3-min cooldown period.

Immediately following the 3-min cooldown period, participants were directed to lie in the supine position for 3 min while performing their assigned breathwork condition (i.e., BB, CS, or CTL). HRV was analyzed at baseline (Pre), 1–3 min postexercise (Post1), and 4–6 min postexercise (Post2). Upon termination of the first recovery protocol measurement period, participants were assisted from the rower to a padded massage/physical therapy table positioned adjacent to the testing area to reduce any noise that may interfere with the measurements being collected. This period, lasting less than 45 s, was the

only lapse in time between the GXT and the beginning of the second supine recovery period (Post 2). Data collected during this transition were discarded, and investigators ensured sufficient time was provided for adequate metrics to be recorded and collected for analysis. HRR was determined by the absolute reduction in heart rate from peak exercise at 1 min (HRR1) and 3 min (HRR3) postexercise, consistent with methodological standards for recovery metrics (31). A schematic overview of the experimental protocol and recovery timeline is presented in Figure 2.

Statistical Analysis

All data were analyzed using IBM SPSS Statistics software (Version 30.0.0.0, IBM Corp., Armonk, NY). Descriptive statistics were calculated for participant characteristics and outcome variables and are presented as means $\pm$ standard deviation for normally distributed data and as median and interquartile range for

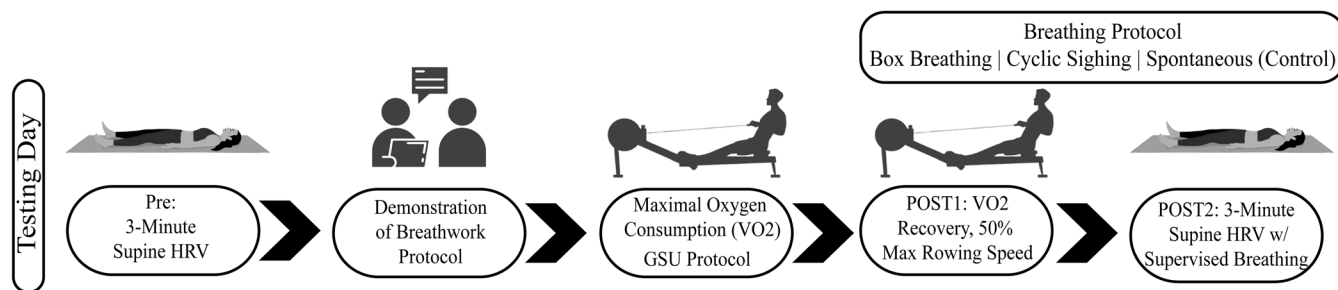


FIGURE 2.—Experimental protocol. Overview of the testing sequence for Reserve Officer Training Corps cadets, including baseline supine HRV assessment (Pre), graded exercise test to volitional exhaustion, 3-min active recovery at 50% of maximal rowing wattage (Post1: 1–3 min postexercise, with the assigned breathing condition: box breathing, cyclic sighing, or spontaneous breathing control), followed by supine recovery HRV measurement (Post2: 4–6 min postexercise). HRR was quantified from peak exercise heart rate at 1 min (HRR1) and 3 min (HRR3) postexercise.

TABLE 1. Participant characteristics.

Characteristics	Total Group (n = 62)	Box Breathing (BB, n = 21)	Cyclic Sighing (CS, n = 21)	Control (CTL, n = 20)	P
Age (yr)	22.0 (3)	20 (3)	20 (3)	20 (3)	0.917
Weight (kg)	77.4 ± 15.4	75.4 ± 9.7	78.6 ± 17.2	78.2 ± 18.6	0.763
Height (cm)	173.1 ± 9.0	172.3 ± 8.4	175.8 ± 10.8	171.1 ± 7.3	0.224
BMI (kg × m ²)	25.7 ± 4.0	25.4 ± 3.4	25.2 ± 3.7	26.5 ± 5.0	0.558
VO _{2max} (mL·kg ⁻¹ ·min ⁻¹)	44.8 ± 10.0	43.1 ± 11.2	47.4 ± 8.78	44.4 ± 9.9	0.265
Sex (% male)	71.0%	61.9%	81.0%	70.0%	0.394

P values for continuous variables derived from one-way ANOVA.

BMI, Body Mass Index.

nonnormally distributed data. Baseline equivalence for age, height, weight, body mass index, and VO_{2max} across all groups were analyzed using a one-way analysis of variance (ANOVA), and chi-square test compared sex distribution. *A priori* power analysis using G*Power 3.1 indicated that a minimum of 42 total participants, ~14 per group, would be required to detect a moderate effect size ($f = 0.25$) with 80% power at an alpha level of 0.05 for the 3 (group) × 3 (time) mixed-design repeated-measures ANOVA. Due to attrition and data quality exclusions, the final analyzed sample comprised 62 participants (BB = 21; CS = 21; CTL = 20), which exceeded the *a priori* minimum. Therefore, the study was adequately powered for detecting moderate effects ($f = 0.25$) at $\alpha = 0.05$.

The primary analysis consisted of a 3 (group: BB, CS, CTL) × 3 (time: baseline, 1–3 min recovery row, 4–6 min post) repeated-measures ANOVA to examine differences in HF-HRV and HRR values across conditions and time points. HF-HRV was derived from R–R interval data exported through Kubios Scientific, using artifact correction at a medium level and smoothness-prior method detrending set at $\lambda = 500$. A 2 (time: HRR1, HRR3) × 3 (group: BB, CS, CTL) repeated-measures ANOVA was used to examine HRR from peak exertion as a marker of recovery from physiological strain. Mauchly's Test of Sphericity was assessed, and Levene's Test of Equality of Error Variance was examined to verify assumptions of homogeneity. Levene's Test indicated unequal variances for HF-HRV at the Post1 timepoint ($P = 0.003$), suggesting heteroskedasticity. Therefore, the Games-Howell post

hoc test was used for between-group pair-wise comparisons. Time-dependent trends were further evaluated using orthogonal contrasts (Linear and quadratic components). Partial eta squared (η_p^2) was reported as a measure of effect size. Statistical significance was set *a priori* at $P \leq 0.05$.

RESULTS

Baseline participant characteristics are presented in Table 1, and descriptive outcome measures for HF-HRV and HRR across all time points and groups are presented in Table 2. Participants reached near-maximal cardiovascular strain during the GXT, achieving a mean peak HR of 189 ± 9 bpm (96%–98% age predicted HR_{max}), with no differences between groups before recovery interventions. Sex-specific descriptives for HRV and HRR outcomes are provided in Supplemental Table 1, Supplemental Digital Content, <https://links.lww.com/MSS/D379>, to support transparency and facilitate future investigations focused on sex-based physiology. The overall sample had a mean age of 21.0 ± 3.0 yr, with comparable means across all groups. Mean body weight was 77.4 ± 15.4 kg, height 173.1 ± 9.0 cm, and body mass index 25.7 ± 4.0 kg·m⁻². A one-way ANOVA indicated no significant differences between groups for age, anthropometric variables such as height, weight, body mass index, VO_{2max}, or sex distribution (all $P > 0.05$).

Heart Rate Variability

A 3 (time: Pre, Post1, Post2) × 3 (group: BB, CS, and CTL) repeated-measures ANOVA revealed a significant

TABLE 2. Descriptive outcomes.

Physiological Measures	Total Group (n = 62)	Box Breathing (BB, n = 21)	Cyclic Sighing (CS, n = 21)	Control (CTL, n = 20)
HF-HRV (ms ²)				
Pre-HF-HRV (ms ²)	2089.2 (3898.7)	2029.6 (2834.6)	2060.3 (3277.9)	2763.6 (3898.6)
Post1 HF-HRV (ms ²)	4.1 (11.1)	8.3 (24.5)	6.0 (49.9)	1.7 (2.7)
Post2 HF-HRV (ms ²)	33.7 (124.0)	61.4 (189.7)	39.8 (216.8)	14.2 (33.3)
Peak HR (bpm)	188.6 ± 9.4	189.0 ± 8.3	188.7 ± 10.3	188.3 ± 10.0
HRR1 (bpm)	130.2 ± 13.4	129.5 ± 12.6	129.3 ± 14.4	132.6 ± 13.5
ΔHRR1 (bpm)	−60.4 (15.5)	−60.5 (15.0)	−61.7 (12.8)	−54.9 (20.7)
HRR3 (bpm)	122.4 ± 16.5	119.9 ± 18.9	123.7 ± 15.0	124.0 ± 15.7
ΔHRR3 (bpm)	−64.9 (16.1)	−63.0 (27.0)	−66.4 (9.8)	−60.0 (17.3)

Data are presented as means ± SD; HRR₁ and HRR₃ represent absolute heart rate (bpm) at 1- and 3-min postexercise; ΔHRR₁ and ΔHRR₃ represent change from peak heart rate (negative = decrease).

main effect for time ($F_{(2,118)} = 465.01$, $P < 0.001$, $\eta^2 = 0.887$) and a significant main effect for group ($F_{(2,59)} = 8.20$, $P < 0.001$, $\eta^2 = 0.217$) when observing HF-HRV. Additionally, a significant time $\times$ group interaction was observed ($F_{(4,118)} = 3.54$, $P = 0.009$, $\eta^2 = 0.207$) such that both breathing intervention (i.e., BB and CS) groups demonstrated significantly greater recovery values compared to spontaneous breathing, when observing parasympathetic activity via HF-HRV (Table 3).

As illustrated in Figure 3, both BB and CS demonstrated substantially greater increases in HF-HRV from Post1 to Post2, compared to that of the CTL group, indicating the acute breathwork interventions produced a stronger recovery response. Orthogonal contrasts indicated a significant linear increase in HF-HRV from Post1 to Post2 for all groups as well as a significant interaction of time and group ($F_{(2,59)} = 5.48$, $P = 0.007$, $\eta^2 = 0.157$). Post hoc comparisons revealed that the CTL group demonstrated significantly lower mean HF-HRV values at both Post1 and Post 2, compared with BB ($P < 0.001$ and $P = 0.002$, respectively) and CS ($P = 0.001$ and $P = 0.002$, respectively). However, no difference was observed between the two breathing interventions ($P = 0.96$).

Heart Rate Recovery

A 2 (time: HRR₁, HRR₃) $\times$ 3 (group: CTL, BB, and CS) repeated-measures ANOVA revealed a significant main effect for time ($F_{(1,59)} = 33.51$, $P < 0.001$, $\eta^2 = 0.362$), indicating that HRR significantly improved from 1 min (HRR₁) to 3 min (HRR₃) postexercise. The group main effect was not significant ($F_{(2,59)} = 0.60$, $P = 0.551$, $\eta^2 = 0.020$) nor was the time $\times$ group interaction ($F_{(2,59)} = 0.95$, $P = 0.391$, $\eta^2 = 0.031$), suggesting that HRR did not differ between BB, CS, or the CTL condition (see Table 3). Overall, HRR improved significantly in all participants over time, independent of breathwork conditions (see Fig. 4).

DISCUSSION

The present study investigated the acute effects of CS and BB compared with spontaneous breathing on autonomic recovery in ROTC cadets following maximal exertion. Significant main effects for time and group, along

with meaningful time $\times$ group interaction, were observed for HF-HRV such that participants who performed structured breathwork were found to have significantly greater HF-HRV at both postexertion timepoints as compared with those performing spontaneous breathing. This indicates that the use of structured breathwork may accelerate parasympathetic reactivation compared with the sole use of spontaneous breathing. To the authors' knowledge, this investigation is the first to examine breathwork interventions and tactical populations immediately following maximal exertion, providing novel insight into how controlled respiratory techniques possibly enhance autonomic recovery during maximal exertion, such as a simulated high-stress operation.

These findings are supported by those of previous studies suggesting that controlled respiration may lead to enhanced vagal tone and autonomic balance through baroreflex stimulatory processes and modulation of respiratory sinus arrhythmia (11–13). Both BB and CS produced increases in HF-HRV from Post1 to Post2, suggesting that when tactical personnel intentionally modulate respiration, even within a 3-min window, meaningful shifts occur with autonomic balance back toward parasympathetic dominance following maximal effort.

Recent work also suggests that BB may have interactions with metabolic load, when compared to other slow breathing techniques paced at 6 breaths per min. Kasap and Aydin (32) observed that BB amplified heart rate immediately after high-intensity interval training likely due to brief breath-holds that elevate carbon dioxide levels and initiate transient sympathetic activation. In contrast, our investigation did not observe any similar adverse effects when comparing BB to CS, as both acute interventions enhanced HF-HRV postexercise. The discrepancy in findings between these similar studies was likely due to recovery positioning, exercise modality, and cadence structures of interventions. This reinforces findings that autonomic effects are highly context-dependent and breathing techniques produce varying results due to individual physiological adaptations.

While breathwork appeared to result in subtle vagal tone modulation changes in the current investigation, represented by changes in HF-HRV, HRR reflected a broader physiological indicator of cardiac recovery. Both metrics represent complementary windows of

TABLE 3. Repeated-measures ANOVA results for HF-HRV and HRR.

Dependent Variable	Source	df	F	P	η^2_p
Log-HF-HRV (ms ²)	Time	2, 118	465.01	<0.001	0.887
	Group	2, 59	8.20	<0.001	0.217
	Time $\times$ group	4, 118	3.54	0.009	0.207
HRR (bpm)	Time	1, 59	33.51	<0.001	0.362
	Group	2, 59	0.60	0.551	0.020
	Time $\times$ group	2, 59	0.95	0.391	0.031

η^2 , partial eta squared.

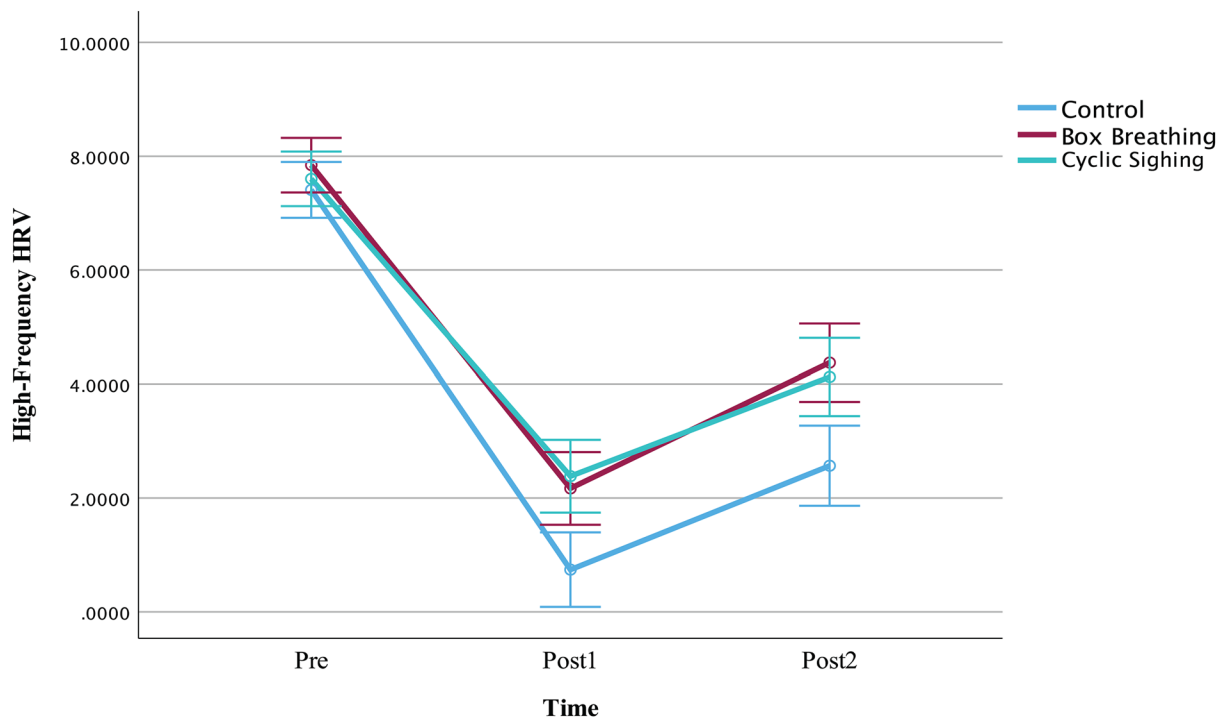


FIGURE 3.—Changes in high-frequency heart rate variability across time points by group. Model-estimated marginal means ($\pm 95\%$ confidence intervals) for log-transformed high-frequency heart rate variability (HF-HRV) measured at baseline (Pre), immediately following exercise (Post1), and following the recovery intervention (Post2) for the box breathing, cyclic sighing, and control groups.

autonomic recovery and parasympathetic reactivation. However, HF-HRV is a more sensitive marker, while HRR appears to integrate a more holistic cardiovascular recovery response. The analysis of both markers of physiological recovery in tandem provides insight as to

whether the influence of structured breathwork on autonomic balance at the neural level and functional recovery can be meaningfully reflected simultaneously, which is an important consideration for translating findings into field-based applications for tactical populations.

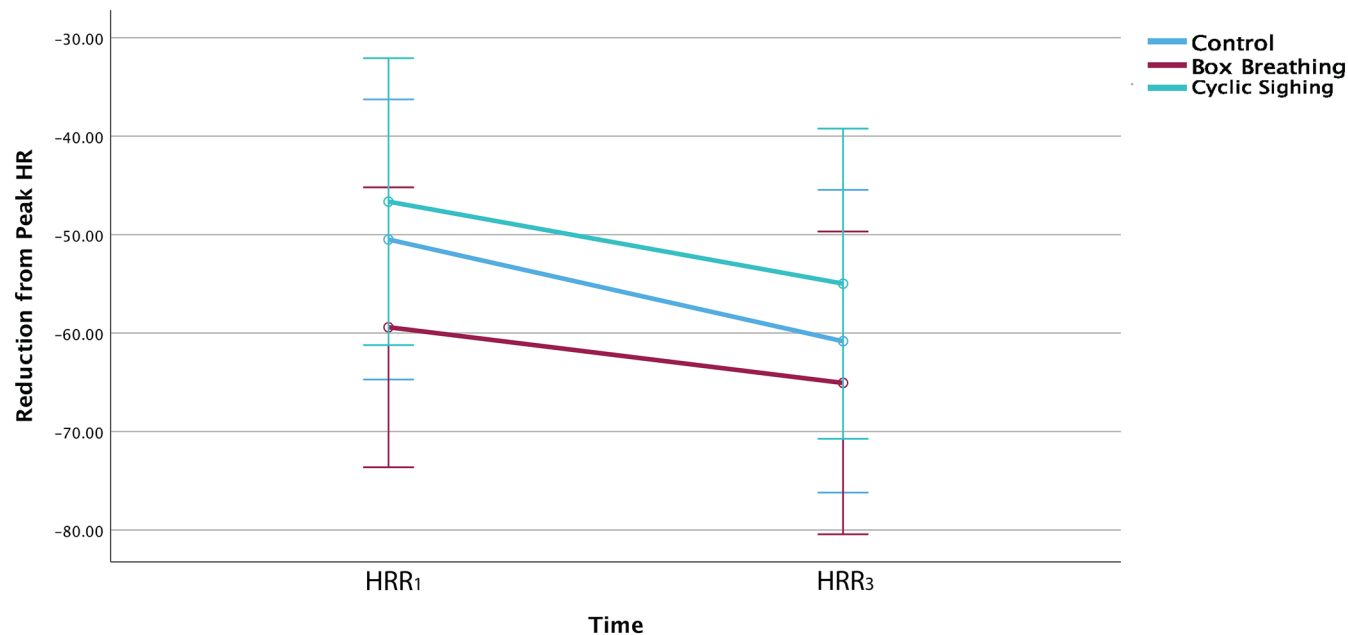


FIGURE 4.—Changes in heart rate recovery across time points by group. Model-estimated marginal means ($\pm 95\%$ confidence intervals) for heart rate recovery at 1 min (ΔHRR_1) and 3 min (ΔHRR_3) postexercise for the box breathing, cyclic sighing, and control groups. Heart rate recovery values represent the change from peak exercise heart rate.

The rapid recovery response of HF-HRV aligns with evidence from current sports science literature indicating that shorter postexercise measurement periods are highly sensitive to vagal reactivation and may provide comparable insight regarding autonomic recovery as compared with longer assessments while offering greater practicality in applied settings. Although Jamieson stated that extended postexercise monitoring periods, such as 5- or 10-min periods, can capture the full trajectory of autonomic normalization, this group reported that the majority of meaningful rebound in the parasympathetic activity occurs within the first 3-min (33). This suggests that short window assessments, such as 3-min measurement periods, may provide sufficient information on recovery, particularly relevant for field-based and tactical applications where efficiency and validity are essential (33). Additionally, findings from the current study provide further evidence in alignment with existing research indicating that rhythmic breathing in a slower cadence of 5–6 breaths per min facilitates a vagally mediated recovery process (30).

The results of the study revealed a large and practically meaningful difference between groups in terms of autonomic recovery. Parasympathetic rebound, as indicated via HF-HRV, occurred robustly with both BB and CS following maximal exertion, compared to spontaneous breathing. The magnitude of effect indicates that a considerable portion of the variation in recovery patterns was driven by acute breathing interventions. Moreover, this represents a physiologically meaningful impact consistent with outcomes reported in structured biofeedback (34) and resonance-frequency breathing (12). Furthermore, evidence from recent controlled studies supports this pattern; Balban et al. (30) suggest that 5 min of daily CS reduced respiratory rate and improved mood in a more effective manner compared with other methods such as mindfulness meditation. This highlights a strong link to parasympathetic recovery via extended exhalation patterns. Rottger et al. (16) demonstrated that BB, rapid autonomic downregulation during high psychological pressure conditions, ultimately reducing physiological arousal while maintaining task performance through periods of cognitive load. Similarly, Ahmed et al. (35) observed measurable increases in pulmonary function, reinforcing its role in the enhancement of cardiorespiratory efficiency. Together, these findings highlight the notion that controlled respirations within the range of 4–6 breaths per min reliably promote reactivation of the parasympathetic nervous system and enhance stability of cardiovascular properties across both clinical and operational contexts for tactical personnel.

From an applied perspective, these results suggest that structured breathwork may produce noticeable recovery benefits within just 3 min after concluding maximally exerted efforts. This duration aligns well with many tactical transitions, specifically in a military context where

it mirrors the short pause following a firefight or movement to contact. Here, Soldiers must rapidly downregulate physiological arousal to maintain motor control and restore focus while preparing for a potential subsequent engagement. For law enforcement and fire service employees, this same period may reflect, respectively, a brief recovery during foot pursuit leading to restraining a subject, or a short interval before reentering a structure after a knockdown. During this time, the focus should be on stabilizing heart rate and focusing on respiration rate to enhance oxygen efficiency, reduce fatigue, and potentially improve precision during high-intensity scenarios. The observed rebound of HF-HRV from Post1 to Post2 in this investigation confirms that the deliberate focus on breathing cadence can accelerate vagal reactivation without the use of any specialized equipment or extending downtime, thereby directly enhancing mission effectiveness, and improving physiological recovery and readiness. Therefore, incorporating BB or CS into a posttraining scenario may offer a scalable and simplistic approach to improve resilience in settings where military and first responders would benefit from enhanced cardiovascular recovery and cognitive readiness.

Although HRR improved significantly across all groups, the lack of between-group differences suggests that HRR may be less responsive to acute respiratory interventions. Our findings support the notion that HRR primarily reflects intrinsic fitness and cardiac properties, rather than a short-term response to vagal tone modulation. Consistent with this interpretation, aerobic fitness did not appear to meaningfully moderate HRR or HF-HRV responses to the breathwork interventions in this present study. In a practical sense, this reinforces the use of HRR for routine monitoring of recovery and cardiorespiratory fitness, considering its reliability and easily measurable process for both field-based and laboratory environments. Conversely, HF-HRV was found to be sensitive to deliberate breathwork interventions, indicating greater utility for a more controlled setting aimed at research-grade precision. Collectively, the results suggest that HF-HRV may serve as a valuable diagnostic tool when monitoring adaptations within the autonomic nervous system, whereas HRR provides a field-expedient method for assessing overall recovery aimed at determining readiness on a moment's notice.

Several limitations should be acknowledged when interpreting the findings of this investigation. Variability in breathwork adherence throughout both Post1 and Post2 measurement periods may have influenced external validity, particularly early in these phases. HRV is particularly influenced by external noise, participant's hydration status, and ongoing psychological stressors (36). Prior practice of breathing techniques among some participants could not be ruled out, potentially affecting comparisons between groups. An additional consideration is the potential influence of time of day on postexercise

autonomic recovery. The participant sessions were scheduled based on availability and were not standardized by the time of day. However, prior work indicates that parasympathetic reactivation following maximal exercise may be attenuated in the evenings compared with morning for HRV dash-derived indices (37). Consistent with this literature, HRR did not differ between groups in the present investigation; however, with unstandardized times of testing HF-HRV responses may have additional variability due to the current design. Furthermore, another consideration relating to female-specific physiology is fluctuations in HRV across the menstrual cycle. This has been reported with reductions in vaguely mediated indices commonly observed during mid to late luteal phase (38). However, with this investigation, menstrual cycle phases were not controlled for or included as a covariant.

Additionally, with the use of a 3×3 repeated-measures ANOVA, the reduced group sizes may have limited statistical power and sensitivity to detect interaction effects between group and time. Mixed model designs are particularly sensitive to sample size imbalances and assumption violations. As such, nonsignificant findings should be interpreted cautiously as subtle physiological differences may not have reached statistical significance due to limited power rather than a true absence of effect (39).

The inclusion of only ROTC cadets versus a sample with more varied military experience limits generalizability within all tactical populations, and the short intervention duration and requirement to perform a newly learned skill under postmaximal-exertion fatigue may have limited full expression of autonomic recovery within the participants. Future investigations in self-selected breathing patterns to determine whether preferences of breathing work carry more weight are warranted. Additionally, the supine position used in the current investigation does not replicate a tactically applicable resting position during real-world scenarios; therefore, a more realistic recovery position, such as “Taking a knee,” should be evaluated against the supine position to evaluate whether similar autonomic recovery and efficacy outcomes are achieved.

CONCLUSIONS

Both BB and CS elicited significantly greater vagal reactivation, as indicated by HF-HRV values, as compared with spontaneous breathing following maximal exertion. These findings suggest that a brief structured breathwork intervention can acutely enhance parasympathetic recovery in tactical populations. This investigation provides further evidence that deliberate respiratory modulation may serve as a powerful systemic regulator bridging the gap between psychological and physiological aspects of recovery. The results support incorporating BB and/or CS into tactical recovery processes to promote

autonomic balance and sustain operational effectiveness among tactical professionals.

The authors would like to thank the ROTC Cadets who volunteered to participate in this study.

All equipment was institutionally owned, and no commercial entities influenced the design, execution or reporting of the study. The authors declare no professional relationships with companies or manufacturers that would benefit from the results of the present study. No financial or commercial conflicts of interest exist.

The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

Data used in this study were reviewed by the Georgia Southern University and Concordia University of Chicago review boards (H26016) and were further approved by the U.S. Army Cadet Command and the Department of Defense Human Research Protections Office (HPRO).

No funding or conflicts of interest were disclosed.

The opinions or assertions contained herein are the private views of the authors and are not to be construed as official or as reflecting the views of the Department of War. Any citations or commercial organizations and trade names in this article do not constitute an official Department of War endorsement or approval of the products or services of these organizations.

REFERENCES

1. Taylor MK, Venkatesh A, Campbell JS, et al. Development and validation of the military readiness scale in U.S. Naval aircrew. *Mil Med*. 2025;190(11-12):e2420–6.
2. Tornero-Aguilera JF, Stergiou M, Rubio-Zarapuz A, Martín-Rodríguez A, Massuça LM, Clemente-Suárez VJ. Optimising combat readiness: practical strategies for integrating physiological and psychological resilience in soldier training. *Healthcare*. 2024;12(12):1160.
3. Thunholm P, Henåker L. A tentative model on effective army combat tactics. *Comp Strategy*. 2020;39(5):490–504.
4. Dong JG. The role of heart rate variability in sports physiology (review). *Exp Ther Med*. 2016;11(5):1531–6.
5. Shaffer F, Ginsberg JP. An overview of heart rate variability metrics and norms. *Front Public Health*. 2017;5:258.
6. Peçanha T, Silva-Júnior ND, de Moraes Forjaz CL. Heart rate recovery: autonomic determinants, methods of assessment and association with mortality and cardiovascular diseases. *Clin Physiol Funct Imaging*. 2014;34(5):327–39.
7. Cole C, Blackstone E, Pashkow F, Snader C, Lauer M. Heart-rate recovery immediately after exercise as a predictor of mortality heart-rate recovery immediately after exercise as a predictor of mortality abstract. *N Engl J Med*. 1999;341(18):1351–7.
8. Tomes C, Schram B, Orr R. Relationships between heart rate variability, occupational performance, and fitness for tactical personnel: a systematic review. *Front Public Health*. 2020;8:583336.
9. Qiu S, Cai X, Sun Z, et al. Heart rate recovery and risk of cardiovascular events and all-cause mortality: a meta-analysis of prospective cohort studies. *J Am Heart Assoc*. 2017;6(5):e005505.
10. Daanen H, Lamberts R, Kallen V, Jin A, Meeteren N. A systematic review on heart-rate recovery to monitor changes in training status in athletes. *Int J Sports Physiol Perform*. 2012;7(3):251–60.
11. Jerath R, Edry JW, Barnes VA, Jerath V. Physiology of long pranayamic breathing: neural respiratory elements may provide a mechanism that explains how slow deep breathing shifts the autonomic nervous system. *Med Hypotheses*. 2006;67(3):566–71.

12. Lehrer P, Kaur K, Sharma A, et al. Heart rate variability biofeedback improves emotional and physical health and performance: a systematic review and meta-analysis. *Appl Psychophysiol Biofeedback*. 2020;45(3):109–29.
13. Zaccaro A, Piarulli A, Laurino M, et al. How breath-control can change your life: a systematic review on psychophysiological correlates of slow breathing. *Front Hum Neurosci*. 2018;12:353.
14. Vlemingx E, Abelson JL, Lehrer PM, Davenport PW, Van Diest I, Van Den Bergh O. Respiratory variability and sighing: a psychophysiological reset model. *Biol Psychol*. 2013;93(1):24–32.
15. Picard M, McEwen BS, Epel ES, Sandi C. An energetic view of stress: focus on mitochondria. *Front Neuroendocrinol*. 2018;49:72–85.
16. Röttger S, Theobald DA, Abendroth J, Jacobsen T. The effectiveness of combat tactical breathing as compared with prolonged exhalation. *Appl Psychophysiol Biofeedback*. 2021;46(1):19–28.
17. Fincham GW, Strauss C, Montero-Marín J, Cavanagh K. Effect of breathwork on stress and mental health: a meta-analysis of randomised-controlled trials. *Sci Rep*. 2023;13(1):432.
18. Letzler R, Janoska-Bedi S, Penner A. Marching into the leadership pipeline: gender, college type, and ROTC participation. *Socius*. 2025;11. doi:10.1177/23780231241297776.
19. Warburton DER, Jamnik VK, Bredin SSD, Gledhill N. The physical activity readiness questionnaire for everyone (PAR-Q+) and electronic physical activity readiness medical examination (ePARmed-X+). *Health Fit J Can*. 2011;4(2):3–17.
20. Bonikowske AR, Christle JW, Gallo PM, Ozemek Cemal. *ACSM's Guidelines for Exercise Testing and Prescription*. 11th ed. Wolters Kluwer; 2025. p. 645.
21. Krummenacher M, Tarvainen M, Montet E, Turner MC, Gu-seva Canu I. Which device is most suitable for measuring heart rate variability in the field? A comparative evaluation of two leading options. *J Occup Environ Med*. 2025;67(9):717–22.
22. Gilgen-Ammann R, Schweizer T, Wyss T. RR interval signal quality of a heart rate monitor and an ECG Holter at rest and during exercise. *Eur J Appl Physiol*. 2019;119(7):1525–32.
23. Schaffarczyk M, Rogers B, Reer R, Gronwald T. Validity of the Polar H10 Sensor for heart rate variability analysis during resting state and incremental exercise in recreational men and women. *Sensors (Basel)*. 2022;22(17):6536.
24. Gabani V, Murugan S. Reliability and validity of kubios HRV smart phone application as measures of heart rate variability. *Int J Curr Res Rev*. 2024;16(06):07–10.
25. Gao WD, Nuuttila OP, Fang HB, Chen Q, Chen X. A new fitness test of estimating VO2max in well-trained rowing athletes. *Front Physiol*. 2021;12:701541.
26. Turnes T, Possamai LT, Penteadó dos Santos R, de Aguiar RA, Ribeiro G, Caputo F. Mechanical power during an incremental test can be estimated from 2000-m rowing ergometer performance. *J Sports Med Phys Fitness*. 2020;60(2):214–9.
27. Penichet-Tomas A, Jimenez-Olmedo JM, Pueo B, Olaya-Cuartero J. Physiological and mechanical responses to a graded exercise test in traditional rowing. *Int J Environ Res Public Health*. 2023;20(4):3664.
28. Montoye AHK, Vondrasek JD, Hancock JB. Validity and reliability of the VO2 Master Pro for oxygen consumption and ventilation assessment. *Int J Exerc Sci*. 2020;13(4):921–32.
29. Gnacinski S. Strategies aimed at optimizing mental recovery from training and occupational performance. *TSAC Rep*. 2021;59(4):18–21.
30. Balban MY, Neri E, Kogon MM, et al. Brief structured respiration practices enhance mood and reduce physiological arousal. *Cell Rep Med*. 2023;4(1):100895.
31. Goldberger JJ, Arora R, Buckley U, Shivkumar K. Autonomic nervous system dysfunction: JACC Focus Seminar. *J Am Coll Cardiol*. 2019;73(10):1189–206.
32. Kasap M, Aydin GR. Box breathing or six breaths per minute: which strategy improves athletes post-HIIT cardiovascular recovery? *PLoS One*. 2025;20(11):e0336615.
33. Jamieson J. Heart rate and heart rate variability. In: French D, Torres R, editors. *NSCA's Essentials of Sport Science*. 1st ed. Human Kinetics; 2022. pp. 391–412.
34. Steffen PR, Austin T, DeBarros A, Brown T. The impact of resonance frequency breathing on measures of heart rate variability, blood pressure, and mood. *Front Public Health*. 2017;5:222.
35. Ahmed A, Devi RG, Priya AJ. Effect of box breathing technique on lung function test. *J Pharm Res Int*. 2021;33(58A):25–31.
36. Damoun N, Amekran Y, Taiek N, El Hangouche AJ. Heart rate variability measurement and influencing factors: towards the standardization of methodology. *Glob Cardiol Sci Pract*. 2024;2024(4):e202435.
37. Brito L, Peçanha T, Tinucci T, Silva-Junior N, Costa L, Forjaz C. Time of day affects heart rate recovery and variability after maximal exercise in pre-hypertensive men. *Chronobiol Int*. 2015;32(10):1385–90.
38. Pestana ER, Mostarda CT, Silva-Filho AC, Salvador EP, de Carvalho WRG. Effect of different phases of menstrual cycle in heart rate variability of physically active women. *Sport Sci Health*. 2018;14(2):297–303.
39. Lakens D. Sample size justification. *Collabra Psychol*. 2022;8(1):33267.